

Project Design Phase

Solution Architecture

Date	15 Feb 2026
Team ID	LTVIP2026TMIDS36802
Project Name	Empowering India: Analysing the Evolution of Union Budget Allocation for Sustainable Growth
Maximum Marks	4 Marks

Solution Architecture :

1. Data Source Layer :

- Collects official Union Budget data for financial years 2021–2024.
- Includes ministry-wise and sector-wise allocation details.
- Covers revenue, capital expenditure, and total budget amounts.
- Data gathered from government reports and open data portals.
- Available in CSV, Excel, and structured table formats for database import.

2. Data Ingestion Layer :

- Extracts raw data from CSV, Excel, and government reports.
- Cleans and pre-processes data (remove duplicates, handle missing values, format columns).
- Transforms data into structured format suitable for database storage.
- Loads processed data into MySQL tables using import tools / SQL commands.
- Ensures data validation and consistency before moving to analysis layer.

3. Data Storage Layer :

- Stores cleaned and structured budget data in MySQL Database.
- Organizes data into tables like Year, Ministry, Sector, and Allocation.
- Maintains relationships using primary keys and foreign keys.
- Ensures data integrity, consistency, and secure storage.
- Provides structured data access for SQL queries and Tableau visualization

4. Visualization & Reporting Layer:

- Connects Tableau to MySQL database for real-time data access.
- Creates interactive dashboards with charts, KPIs, and filters (Year, Ministry, Sector).
- Visualizes trends, comparisons, and budget distribution clearly.
- Generates summary reports and downloadable insights.
- Helps users analyze data and support academic or business decision-making.

Example :

- A Finance Analyst wants to understand why Healthcare allocation increased significantly in 2024
- The dashboard shows a 15% rise compared to 2023.
- Sector comparison chart highlights increased government focus on public health programs.
- Revenue vs Capital chart shows higher capital investment in hospitals and infrastructure.
- Trend line clearly indicates steady growth from 2021 to 2024

As a result, policymakers use this insight to justify increased healthcare funding and plan future investments effectively — all through real-time interactive dashboards

